

Sentiment Analysis: Corpus-based approaches

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Text Processing

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Learning objectives

- ▶ Explain the Naïve Bayes model for sentiment analysis
- ▶ Explain the Laplace smoothing technique
- ▶ Explain the Binary Naïve Bayes model
- ▶ Identify other tasks for which the same approach is applicable

Sentiment Analysis: 2 main approaches

- ▶ **Lexicon based**

- ▶ Binary
- ▶ Graggable

- ▶ **Corpus based**

- ▶ **Naive Bayes**
- ▶ Deep Learning

Corpus-based approaches: Naive Bayes

Principle

Assign the **sentiment** or **class** having the highest **posterior probability**.

Namely, determine the sentiment s^* of text T such that:

$$s^* = \underset{s_i}{\operatorname{argmax}} p(s_i | T) \text{ for } s_i \in \{\text{negative}, \text{positive}, \text{neutral}\}$$

$p(s_i | T)$ cannot be directly estimated...

Corpus-based approaches: Naive Bayes

Principle

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$p(s_i | T)$ cannot be directly estimated...

→ use the **Bayes rule**:

$$p(s_i | T) = \frac{p(T | s_i)p(s_i)}{p(T)}$$

Corpus-based approaches: Naive Bayes

Bayes classifier:

$$\mathbf{s}^* = \operatorname{argmax}_{s_i} \frac{p(T|s_i)p(s_i)}{p(T)}$$

Corpus-based approaches: Naive Bayes

Bayes classifier:

$$\mathbf{s}^* = \operatorname{argmax}_{s_i} \frac{p(T|s_i)p(s_i)}{\mathbf{p(T)}}$$

Since **evidence** $\mathbf{p(T)}$ is independent of s_i , we can ignore it

$$\mathbf{s}^* = \operatorname{argmax}_{s_i} p(T|s_i)p(s_i)$$

Corpus-based approaches: Naive Bayes

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Since **evidence** $p(T)$ is independent of s_i , we can ignore it

$$\mathbf{s}^* = \operatorname{argmax}_{s_i} p(T|s_i)p(s_i)$$

- ▶ $p(s_i|T)$ is the **posterior probability**
- ▶ $p(T|s_i)$ is the **likelihood**
- ▶ $p(s_i)$ is the **prior probability**

Corpus-based approaches: Naive Bayes

How to compute the likelihood?

Assume that T is described by a number of **features** or attributes $t_1, t_2, \dots, t_N$

Corpus-based approaches: Naive Bayes

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Naive assumption: **features** are **independent**

$$p(T|s_i) = p(t_1, t_2, \dots, t_N|s_i) \approx \prod_{j=1}^N p(t_j|s_i)$$

Corpus-based approaches: Naive Bayes

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Naive assumption: **features** are **independent**

$$p(T|s_i) = p(t_1, t_2, \dots, t_N|s_i) \approx \prod_{j=1}^N p(t_j|s_i)$$

$\Rightarrow$ product of probabilities of each **feature** value of text occurring with class s_i

Corpus-based approaches: Naive Bayes

How to compute the prior probability?

→ corresponds to the safest decision when no other information is given $\sim$ majority voting

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Requires an **annotated corpus** (text along with their sentiment)

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How to compute the prior probability?

→ corresponds to the safest decision when no other information is given $\sim$ majority voting

Requires an **annotated corpus** (text along with their sentiment)

Compute **prior probability** by simple relative frequency

$$p(s_i) = \frac{\text{count}(s_i)}{\sum_{j=0}^J \text{count}(s_j)}$$

with J the number of different classes and $\text{count}(\cdot)$ is the counting function

Digression: corpus-based machine learning

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- ▶ **development** set, also called **validation** set → used to **design** the model
 - ▶ e.g. feature selection, set meta-parameters (could be some weights)
 - ▶ ground truth (correct labels) available
 - ▶ used to select the best model

Digression: corpus-based machine learning

- ▶ **training** set → used to **estimate probabilities**
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 - ▶ e.g. feature selection, set meta-parameters (could be some weights)
 - ▶ ground truth (correct labels) available
 - ▶ used to select the best model
- ▶ **test** set → used to **evaluate generalisation** power
 - ▶ unseen examples
 - ▶ best case: no access to the ground truth

Naive Bayes classifier

Final decision

$$\mathbf{s}^* = \operatorname{argmax}_{s_i} p(s_i) \prod_{j=1}^N p(t_j | s_i)$$

Naive Bayes classifier

Final decision

$$\mathbf{s}^* = \operatorname{argmax}_{s_i} p(s_i) \prod_{j=1}^N p(t_j | s_i)$$

1. Compute **prior probability** of each class

Naive Bayes classifier

Final decision

$$\mathbf{s}^* = \operatorname{argmax}_{s_i} p(s_i) \prod_{j=1}^N p(t_j | s_i)$$

1. Compute **prior probability** of each class
2. For each class:
 - ▶ Compute **likelihood** of each feature

Naive Bayes classifier

Final decision

$$\mathbf{s}^* = \underset{s_i}{\operatorname{argmax}} p(s_i) \prod_{j=1}^N p(t_j | s_i)$$

1. Compute **prior probability** of each class
2. For each class:
 - ▶ Compute **likelihood** of each feature
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Naive Bayes classifier

Final decision

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1. Compute **prior probability** of each class
2. For each class:
 - ▶ Compute **likelihood** of each feature
3. Calculate the **posterior probability** by product of previous components
4. Select sentiment having maximum posterior probability
→ **negative**, **positive** or **neutral**

Naive Bayes for Sentiment Analysis - Example

Consider the following dummy **training** corpus of 7 movie reviews:

Doc	Words	Class
1	great movie, excellent plot, renowned actors	positive
2	I had not seen a fantastic plot like this in good 5 years. amazing!!!	positive
3	lovely plot, amazing cast, somehow I am in love with the bad guy	positive
4	bad movie with great cast, but very poor plot and unimaginative ending	negative
5	I hate this film, it has nothing original	negative
6	great movie, but not...	negative
7	very bad movie, I have no words to express how I dislike it	negative

Naive Bayes for Sentiment Analysis - Example

Compute **prior probability** of each class by relative frequency

$$p(\text{positive}) = \frac{\text{count}(\text{positive})}{\sum_{s \in \{\text{positive}, \text{negative}\}} \text{count}(s)} = \frac{3}{7} = 0.43$$

$$p(\text{negative}) = \frac{\text{count}(\text{negative})}{\sum_{s \in \{\text{positive}, \text{negative}\}} \text{count}(s)} = \frac{4}{7} = 0.57$$

Naive Bayes for Sentiment Analysis - Example

What **features** should we consider?

- ▶ could use **all** words
 - ▶ but some might not be relevant → we are interested in the **emotion words**
- ▶ in this example: focus on **adjectives** (**bag-of-word** representation)

Doc	Words	Class
1	great movie, excellent plot, renowned actors	positive
2	I had not seen a fantastic plot like this in good 5 years. amazing !!!	positive
3	Lovely plot, amazing cast, somehow I am in love with the bad guy	positive
4	bad movie with great cast, but very poor plot and unimaginative ending	negative
5	I hate this film, it has nothing original	negative
6	great movie, but bad casting...	negative
7	Very bad movie, I have no words to express how I dislike it	negative

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What **features** should we consider?

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 - ▶ but some might not be relevant → we are interested in the **emotion words**
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Doc	Words	Class
1	great excellent renowned	positive
2	fantastic good amazing !!!	positive
3	lovely amazing bad	positive
4	bad great poor unimaginative	negative
5	original	negative
6	great bad	negative
7	bad	negative

Naive Bayes for Sentiment Analysis - Example

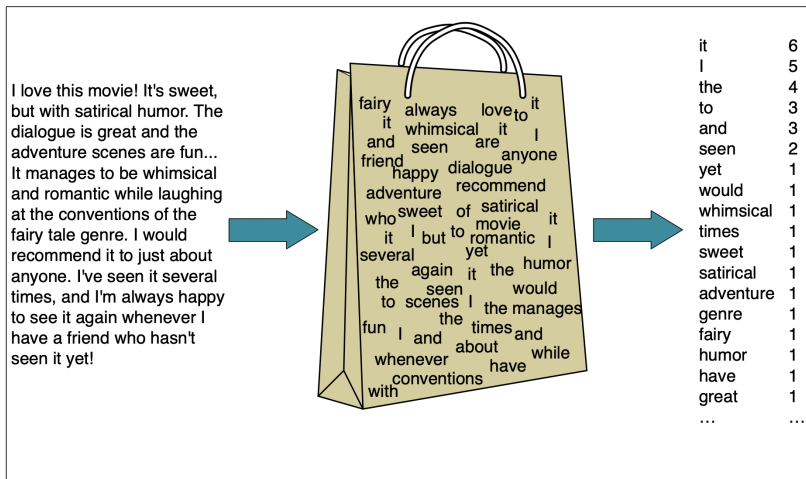


Figure 4.1 Intuition of the multinomial naive Bayes classifier applied to a movie review. The position of the words is ignored (the *bag of words* assumption) and we make use of the frequency of each word.

[Jurafsky and Martin, 2021]

Naive Bayes for Sentiment Analysis - Example

Compute the **likelihoods** for all features in a given each class

Important

Assume **standard pre-processing**: tokenisation, lowercasing, punctuation removal (but keep special punctuation, e.g. "!!!")

Examples:

- ▶ GOOD = GooD = Good = good
- ▶ I'll = I will (though not relevant here)
- ▶ aren't = are not

Naive Bayes for Sentiment Analysis - Example

Compute the **likelihoods** for all features in a given each class

$$p(t_j | s_i) = \frac{\text{count}(t_j, s_i)}{\sum_f \text{count}(t_f, s_i)} \rightarrow \text{relative frequency}$$

p(amazing	positive)	= 2/10
p(bad	positive)	= 1/10
p(excellent	positive)	= 1/10
p(fantastic	positive)	= 1/10
p(good	positive)	= 1/10
p(great	positive)	= 1/10
p(love	positive)	= 1/10
p(original	positive)	= 0/10
p(poor	positive)	= 0/10
p(renowned	positive)	= 1/10
p(unimaginative	positive)	= 0/10
p(!!!	positive)	= 1/10

p(amazing	negative)	= 0/8
p(bad	negative)	= 3/8
p(excellent	negative)	= 0/8
p(fantastic	negative)	= 0/8
p(good	negative)	= 0/8
p(great	negative)	= 2/8
p(love	negative)	= 0/8
p(original	negative)	= 1/8
p(poor	negative)	= 1/8
p(renowned	negative)	= 0/8
p(unimaginative	negative)	= 1/8
p(!!!	negative)	= 0/8

Naive Bayes for Sentiment Analysis - Example

Relative frequencies for prior and likelihoods make the model in a Naive Bayes classifier

→ features are supposed to be **independent** (no covariance taken into account)

→ this is an approximation of course

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→ the set of all prior probabilities and likelihoods

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Relative frequencies for prior and likelihoods make the model in a Naive Bayes classifier

→ features are supposed to be **independent** (no covariance taken into account)

→ this is an approximation of course

What is the model?

→ the set of all prior probabilities and likelihoods

At **test** time, this model is used to find the most likely class (sentiment) for the unknown text

$$\mathbf{s}^* = \underset{s_i}{\operatorname{argmax}} p(s_i) \prod_{j=1}^N p(t_j | s_i)$$

Naive Bayes for Sentiment Analysis - Text Ex. 1

Consider the following **test segment** to classify:

Doc	Words	Class
8	This was a fantastic story, great, lovely	?

Naive Bayes for Sentiment Analysis - Text Ex. 1

Consider the following **test segment** to classify:

1. Extract features

Doc	Words	Class
8	This was a fantastic story, great , lovely	?

Naive Bayes for Sentiment Analysis - Text Ex. 1

Consider the following **test segment** to classify:

1. Extract features
2. Build representation

Doc	Words	Class
8	fantastic great lovely	?

Naive Bayes for Sentiment Analysis - Text Ex. 1

Consider the following **test segment** to classify:

1. Extract features
2. Build representation

Doc	Words	Class
8	fantastic great lovely	?

3. Get **likelihoods**

$$\begin{array}{lcl} p(\text{fantastic} | \text{positive}) & = & 1/10 \\ p(\text{great} | \text{positive}) & = & 1/10 \\ p(\text{lovely} | \text{positive}) & = & 1/10 \end{array} \quad \left| \quad \begin{array}{lcl} p(\text{fantastic} | \text{negative}) & = & 0/8 \\ p(\text{great} | \text{negative}) & = & 2/8 \\ p(\text{lovely} | \text{negative}) & = & 0/8 \end{array}$$

Naive Bayes for Sentiment Analysis - Text Ex. 1

$$\begin{aligned} p(\text{positive}|\text{text}) &= p(\text{positive}) * p(\text{fantastic}|\text{positive}) * p(\text{great}|\text{positive}) * p(\text{lovely}|\text{positive}) \\ &= 3/7 * 1/10 * 1/10 * 1/10 \\ &= 0.00043 \end{aligned}$$

$$\begin{aligned} p(\text{negative}|\text{text}) &= p(\text{negative}) * p(\text{fantastic}|\text{negative}) * p(\text{great}|\text{negative}) * p(\text{lovely}|\text{negative}) \\ &= 4/7 * 0/8 * 2/8 * 0/8 \\ &= 0 \end{aligned}$$

Final decision: sentiment is **positive**

Naive Bayes for Sentiment Analysis - Text Ex. 2

Doc	Words	Class
1	great movie, excellent plot, renowned actors	positive
2	I had not seen a fantastic plot like this in good 5 years. amazing!!!	positive
3	lovely plot, amazing cast, somehow I am in love with the bad guy	positive
4	bad movie with great cast, but very poor plot and unimaginative ending	negative
5	I hate this film, it has nothing original	negative
6	great movie, but not...	negative
7	very bad movie, I have no words to express how I dislike it	negative

Doc	Words	Class
1	great excellent renowned	positive
2	fantastic good amazing !!!	positive
3	lovely amazing bad	positive
4	bad great poor unimaginative	negative
5	original	negative
6	great bad	negative
7	bad	negative

Consider the following **test segment** to classify:

Doc	Words	Class
9	Great great great	?

Naive Bayes for Sentiment Analysis - Text Ex. 2

Consider the following **test segment** to classify:

Doc	Words	Class
9	Great plot, great cast, great everything	?

Naive Bayes for Sentiment Analysis - Text Ex. 2

Consider the following **test segment** to classify:

1. Extract features
2. Build representation

Doc	Words	Class
9	Great great great	?

Naive Bayes for Sentiment Analysis - Text Ex. 2

Consider the following **test segment** to classify:

1. Extract features
2. Build representation

Doc	Words	Class
9	Great great great	?

3. Get likelihoods:

$$p(\text{great}|\text{positive}) = 1/10 \mid p(\text{great}|\text{negative}) = 2/8$$

Naive Bayes for Sentiment Analysis - Text Ex. 2

Consider the following **test segment** to classify:

1. Extract features
2. Build representation

Doc	Words	Class
9	Great great great	?

3. Get likelihoods:

$$p(\text{great}|\text{positive}) = 1/10 \mid p(\text{great}|\text{negative}) = 2/8$$

4. Compute posteriors

$$p(\text{positive}|\text{text}) = 3/7 * 1/10 * 1/10 * 1/10 = \mathbf{0.00043}$$

$$p(\text{negative}|\text{text}) = 4/7 * 2/8 * 2/8 * 2/8 = \mathbf{0.00893}$$

Final decision: sentiment is **negative**

Naive Bayes for Sentiment Analysis - Text Ex. 2

Consider the following **test segment** to classify:

1. Extract features
2. Build representation

Doc	Words	Class
9	Great gr	?

**Training data
should be
representative!**

3. Get likelihoods.

$$p(\text{great}|\text{positive}) = 1/10 \mid p(\text{great}|\text{negative}) = 2/8$$

4. Compute posteriors

$$p(\text{positive}|\text{text}) = 3/7 * 1/10 * 1/10 * 1/10 = \mathbf{0.00043}$$

$$p(\text{negative}|\text{text}) = 4/7 * 2/8 * 2/8 * 2/8 = \mathbf{0.00893}$$

Final decision: sentiment is **negative**

Naive Bayes for Sentiment Analysis - Text Ex. 3

Doc	Words	Class
1	great movie, excellent plot, renowned actors	positive
2	I had not seen a fantastic plot like this in good 5 years. amazing!!!	positive
3	lovely plot, amazing cast, somehow I am in love with the bad guy	positive
4	bad movie with great cast, but very poor plot and unimaginative ending	negative
5	I hate this film, it has nothing original	negative
6	great movie, but not...	negative
7	very bad movie, I have no words to express how I dislike it	negative

Doc	Words	Class
1	great excellent renowned	positive
2	fantastic good amazing !!!	positive
3	lovely amazing bad	positive
4	bad great poor unimaginative	negative
5	original	negative
6	great bad	negative
7	bad	negative

Consider the following **test segment** to classify:

Doc	Words	Class
10	Excellent cast, unimaginative ending	?

Naive Bayes for Sentiment Analysis - Text Ex. 3

Consider the following **test segment** to classify:

Doc	Words	Class
10	Excellent cast, unimaginative ending	?

Naive Bayes for Sentiment Analysis - Text Ex. 3

Consider the following **test segment** to classify:

1. Extract features
2. Build representation

Doc	Words	Class
10	Excellent unimaginative	?

Naive Bayes for Sentiment Analysis - Text Ex. 3

Consider the following **test segment** to classify:

1. Extract features
2. Build representation

Doc	Words	Class
10	Excellent unimagative	?

3. Get likelihoods:

$$\begin{array}{lcl} p(\text{excellent} | \text{positive}) & = 1/10 & | \quad p(\text{excellent} | \text{negative}) = 0/8 \\ p(\text{unimagative} | \text{positive}) & = 0/10 & | \quad p(\text{unimagative} | \text{negative}) = 1/8 \end{array}$$

Naive Bayes for Sentiment Analysis - Text Ex. 3

Consider the following **test segment** to classify:

1. Extract features
2. Build representation

Doc	Words	Class
10	Excellent unimagative	?

3. Get likelihoods:

$$\begin{array}{lcl} p(\text{excellent}|\text{positive}) = 1/10 & | & p(\text{excellent}|\text{negative}) = 0/8 \\ p(\text{unimagative}|\text{positive}) = 0/10 & | & p(\text{unimagative}|\text{negative}) = 1/8 \end{array}$$

4. Compute posteriors

$$p(\text{positive}|\text{text}) = 3/7 * 1/10 * 0/10 = \mathbf{0.0}$$

$$p(\text{negative}|\text{text}) = 4/7 * 0/8 * 1/8 = \mathbf{0.0}$$

Final decision: sentiment is ???

Naive Bayes for Sentiment Analysis - Text Ex. 3

Consider the following **test segment** to classify:

1. Extract features

2. Build representation

Doc	Words	Class
10	Excellent unimaginative	?

**Training data
should be large!**

3. Get likelihoods:

$$\begin{array}{lcl} p(\text{excellent}|\text{positive}) = 1/10 & | & p(\text{excellent}|\text{negative}) = 0/8 \\ p(\text{unimaginative}|\text{positive}) = 0/10 & | & p(\text{unimaginative}|\text{negative}) = 1/8 \end{array}$$

4. Compute posteriors

$$p(\text{positive}|\text{text}) = 3/7 * 1/10 * 0/10 = \mathbf{0.0}$$

$$p(\text{negative}|\text{text}) = 4/7 * 0/8 * 1/8 = \mathbf{0.0}$$

Final decision: sentiment is ???

Exercise

Doc	Words	Class
1	A <u>sensitive</u> , <u>moving</u> , <u>brilliant</u> work	Positive
2	An <u>edgy</u> thriller that delivers a <u>surprising</u> punch	Positive
3	A <u>sensitive</u> , <u>insightful</u> , <u>flamboyant</u> film	Positive
4	Neither <u>revelatory</u> nor truly <u>edgy</u> – merely crassly <u>flamboyant</u> and comedically <u>labored</u>	Negative
5	<u>Unlikable</u> , <u>uninteresting</u> , <u>unfunny</u> , and completely, utterly <u>inept</u>	Negative
6	A sometimes <u>incisive</u> and <u>sensitive</u> portrait, targeting <u>sensitive</u> topics, that is undercut by its <u>awkward</u> structure	Negative
7	It's a sometimes <u>interesting</u> remake that doesn't compare to the <u>brilliant</u> original	Negative

Classify the following sentence using a Naive Bayes classifier based on the above training data (without smoothing).

Doc	Words	Class
8	A <u>flamboyant</u> romcom, <u>sensitive</u> but <u>awkward</u> at times.	???

Naive Bayes for Sentiment Analysis

We cannot ensure that all possible words appear in the training corpus

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→ apply a **smoothing** technique

$$p(t_j|s_i) = \frac{\text{count}(t_j, s_i) + 1}{\sum_f (\text{count}(t_f, s_i) + 1)}$$

Naive Bayes for Sentiment Analysis

We cannot ensure that all possible words appear in the training corpus

→ apply a **smoothing** technique

$$p(t_j|s_i) = \frac{\text{count}(t_j, s_i) + 1}{(\sum_f \text{count}(t_f, s_i)) + |\mathbf{V}|}$$

→ **Laplace** smoothing, also called **add-1** smoothing

where $|\mathbf{V}|$ is the number of **distinct features** or **types** (also called **vocabulary**)

→ **12 in our example**

great, excellent, renowned, fantastic, good, amazing, !!!, lovely, bad, poor, unimaginative, original

Naive Bayes for Sentiment Analysis - Text Ex. 3

Consider the following **test segment** to classify:

1. Extract features
2. Build representation

Doc	Words	Class
10	Excellent unimaginative	?

Naive Bayes for Sentiment Analysis - Text Ex. 3

Consider the following **test segment** to classify:

1. Extract features
2. Build representation

Doc	Words	Class
10	Excellent unimaginative	?

3. Get new likelihoods:

$$\begin{array}{lcl} p(\text{excellent}|\text{positive}) & = (1+1)/(10+12) = 2/22 & \left| \quad p(\text{excellent}|\text{negative}) = (0+1)/(8+12) = 1/20 \right. \\ p(\text{unimaginative}|\text{positive}) & = (0+1)/(10+12) = 1/22 & \left. p(\text{unimaginative}|\text{negative}) = (1+1)/(8+12) = 2/20 \right. \end{array}$$

Naive Bayes for Sentiment Analysis - Text Ex. 3

Consider the following **test segment** to classify:

1. Extract features
2. Build representation

Doc	Words	Class
10	Excellent unimagative	?

3. Get new likelihoods:

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4. Compute posteriors

$$p(\text{positive}|\text{text}) = 3/7 * 2/22 * 1/22 = \mathbf{0.00176}$$

$$p(\text{negative}|\text{text}) = 4/7 * 1/20 * 2/20 = \mathbf{0.00286}$$

Final decision: sentiment is **negative**

Exercise

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1	A <u>sensitive</u> , <u>moving</u> , <u>brilliant</u> work	Positive
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6	A sometimes <u>incisive</u> and <u>sensitive</u> portrait, targeting <u>sensitive</u> topics, that is undercut by its <u>awkward</u> structure	Negative
7	It's a sometimes <u>interesting</u> remake that doesn't compare to the <u>brilliant</u> original	Negative

Classify the following sentence using a Naive Bayes classifier based on the above training data **with smoothing: Laplace (add-1)**.

Doc	Words	Class
8	A <u>flamboyant</u> romcom, <u>sensitive</u> but <u>awkward</u> at times.	???

Naive Bayes for Sentiment Analysis - Questions

Is this a good solution?

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What is the role of the prior?

- ▶ reminder: safest decision **when no other information is given** $\sim$ majority voting
- ▶ important especially on biased cases

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- ▶ important especially on biased cases

Can we extend to a non-binary classification?

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Is this a good solution?

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Is it robust?

- ▶ Problem: what about new words?

What is the role of the prior?

- ▶ reminder: safest decision **when no other information is given** $\sim$ majority voting
- ▶ important especially on biased cases

Can we extend to a non-binary classification?

- ▶ Naive Bayes can be easily extended by considering more than 2 classes
- ▶ ... but beware of the **curse of dimensionality** $\rightarrow$ sparsity

Naive Bayes for Sentiment Analysis - Questions

How can we improve this solution?

1. Consider other **features**?

- ▶ using all words in Naive Bayes works well for some tasks
- ▶ subsets of words may help → use the **development** set to decide
- ▶ previous examples consider only adjectives, this is limiting
 - ▶ **verbs**: hate, dislike
 - ▶ **intensifiers**: very, much, a lot
 - ▶ **negation**: not ← **very important!**
 - *didn't like this movie*
 - *didn't NOT_like NOT_this NOT_movie*
 - ▶ **nouns**: love, creativity
 - possibly people tend to mostly talk of those **nouns** in a **positive** or **negative** way
- ▶ pre-built polarity lexicons can be helpful

Naive Bayes for Sentiment Analysis - Questions

How can we improve this solution?

1. Consider other **features**?

- ▶ **pre-built polarity lexicons** can be helpful
 - add a feature that is counted whenever a word from a lexicon occurs
 - “word appears in positive/negative lexicon”
 - simple approach → can be effective when the data is sparse

Naive Bayes for Sentiment Analysis - Questions

How can we improve this solution?

2. Consider other **feature representations**?

- ▶ Instead of frequencies → binary count
- ▶ 1 if feature appears in a segment, 0 otherwise

	NB Counts		Binary Counts	
	+	-	+	-
Four original documents:				
- it was pathetic the worst part was the boxing scenes	and	2	0	1
	boxing	0	1	0
	film	1	0	1
- no plot twists or great scenes	great	3	1	2
+ and satire and great plot twists	it	0	1	0
+ great scenes great film	no	0	1	0
	or	0	1	0
	part	0	1	0
	pathetic	0	1	0
	plot	1	1	1
	satire	1	0	1
	scenes	1	2	1
After per-document binarization:	the	0	2	0
- it was pathetic the worst part boxing scenes	twists	1	1	1
- no plot twists or great scenes	was	0	2	0
+ and satire great plot twists	worst	0	1	0
+ great scenes film				

Figure 4.3 An example of binarization for the binary naive Bayes algorithm.

[Jurafsky and Martin, 2021]

Naive Bayes for Sentiment Analysis - Binary example

Consider the following **training set**:

doc	“good”	“poor”	“great”	(class)
d1.	3	0	3	pos
d2.	0	1	2	pos
d3.	1	3	0	neg
d4.	1	5	2	neg
d5.	0	2	0	neg

[Jurafsky and Martin, 2021]

Predict the sentiment of the text:

”A **good**, **good** plot and **great** characters, but **poor** acting.”

Naive Bayes for Sentiment Analysis - Binary example

Consider the following **training set**:

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d5.	0	2	0	neg

[Jurafsky and Martin, 2021]

Predict the sentiment of the text:

”A **good**, **good** plot and **great** characters, but **poor** acting.”

Standard NB:

$$\begin{array}{l|l} p(\text{good}|\text{positive}) = 3/9 = 1/3 & p(\text{good}|\text{negative}) = 1/14 \\ p(\text{great}|\text{positive}) = 5/9 & p(\text{great}|\text{negative}) = 2/14 = 1/7 \end{array}$$

Naive Bayes for Sentiment Analysis - Binary example

Consider the following **training set**:

doc	“good”	“poor”	“great”	(class)
d1.	3	0	3	pos
d2.	0	1	2	pos
d3.	1	3	0	neg
d4.	1	5	2	neg
d5.	0	2	0	neg

[Jurafsky and Martin, 2021]

Predict the sentiment of the text:

“A **good**, **good** plot and **great** characters, but **poor** acting.”

Binary NB:

$$\begin{array}{l|l} p(\text{good}|\text{positive}) = 1/4 & p(\text{good}|\text{negative}) = 2/6 = 1/3 \\ p(\text{great}|\text{positive}) = 2/4 = 1/2 & p(\text{great}|\text{negative}) = 1/6 \end{array}$$

Exercise

Doc	Words	Class
1	A <u>sensitive</u> , <u>moving</u> , <u>brilliant</u> work	Positive
2	An <u>edgy</u> thriller that delivers a <u>surprising</u> punch	Positive
3	A <u>sensitive</u> , <u>insightful</u> , <u>flamboyant</u> film	Positive
4	Neither <u>revelatory</u> nor truly <u>edgy</u> – merely crassly <u>flamboyant</u> and comedically <u>labored</u>	Negative
5	<u>Unlikable</u> , <u>uninteresting</u> , <u>unfunny</u> , and completely, utterly <u>inept</u>	Negative
6	A sometimes <u>incisive</u> and <u>sensitive</u> portrait, targeting <u>sensitive</u> topics, that is undercut by its <u>awkward</u> structure	Negative
7	It's a sometimes <u>interesting</u> remake that doesn't compare to the <u>brilliant</u> original	Negative

Classify the following sentence using a **Binary Naive Bayes** classifier based on the above training data **without smoothing**.

Doc	Words	Class
8	A <u>flamboyant</u> romcom, <u>sensitive</u> but <u>awkward</u> at times.	???

Naive Bayes for Sentiment Analysis - Questions

How can we improve this solution?

3. Consider other **algorithms**?

- ▶ Maximum Entropy (MaxEnt), Support Vector Machines (SVM), neural networks
- no assumption of statistical independence among features
- more complex but tend to do better

→ Take a look at Chapter 5 of [Jurafsky and Martin, 2021] for **Text Classification with Logistic Regression**

Learning objectives

- ▶ Explain the Naïve Bayes model for sentiment analysis
- ▶ Explain the Laplace smoothing technique
- ▶ Explain the Binary Naïve Bayes model
- ▶ Identify other tasks for which the same approach is applicable

Extra reading (optional)

Geriska Isabelle, Warih Maharani and Ibnu Asror (2018).
**Analysis on Opinion Mining Using Combining Lexicon-Based
Method and Multinomial Naive Bayes.**

International Conference on Industrial Enterprise and System
Engineering

Hanhoon Kang, Seong Joon Yoo and Dongil Han (2012).
**Senti-lexicon and improved Naïve Bayes algorithms for
sentiment analysis of restaurant reviews.**

Expert Systems with Applications, v.39, issue 5.

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S0957417411016538](https://www.sciencedirect.com/science/article/pii/S0957417411016538)

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Jurafsky, D. and Martin, J. H. (2021).

Speech and Language Processing, see chapter 4: *Naive Bayes and Sentiment Classification*.